

Near-Perfect Session Retrieval on LongMemEval:

MemoryKG’s Hybrid Semantic-Graph Architecture
Achieves 99.2% Recall@10 and 100% Recall@30 Across 500 Questions

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Abstract

We evaluate MemoryKG, a conversational memory system based on DocKG [Suchanek, 2026b], on the LongMemEval benchmark [Wang et al., 2024], a dataset of 500 questions spanning six question types that require locating specific sessions within a multi-session conversation corpus of up to 500 sessions. MemoryKG achieves **99.2% session-level Recall@10** and **100% Recall@30** across all question types, with an NDCG@10 of 0.954, on the full 500-question test set. Perfect recall is attained at $k=30$ with no LLM required at any stage — neither for indexing nor for retrieval. The hardest question types are *temporal-reasoning* (Recall@10 = 97.7%) and *multi-session* (Recall@10 = 99.2%), where evidence spans multiple chronologically ordered sessions. The remaining four types — *knowledge-update*, *single-session-user*, *single-session-assistant*, and *single-session-preference* — each reach 100% Recall@10. Graph expansion through structured CONTAINS, NEXT, and REFERENCES edges closes the gap between semantic seeding and complete evidence recovery, requiring only $k=30$ seeds to exhaust all 500 questions. A single shared sentence-transformer embedder (BAAI/bge-small-en-v1.5) answers each question in approximately 0.68 seconds on Apple Silicon against a prebuilt index over 19,195 sessions.

Note on this revision. The figures here supersede the April 2026 draft, which reported 99.4% Recall@10 and an NDCG@10 of 0.943. Those came from a code state that predated an unconditional sibling-clustering step in the multi-session reranker; they were not reproducible against the released system. Every number below is from a single re-run on 2026-08-26 at commit 9754508, recorded in `benchmarks/RESULTS_SUMMARY.md`.

1 Introduction

Long-term conversational memory retrieval — locating the right session from a large multi-session corpus given a natural-language question — is increasingly important for AI assistants operating over months of interaction history. Prior approaches include full-context LLM inference, structured memory extraction [Mem0, 2024], and flat dense retrieval over session text. These approaches differ in accuracy, latency, cost, and privacy guarantees.

The LongMemEval benchmark [Wang et al., 2024] provides a structured evaluation across six question types, each requiring the system to locate the correct session(s) from a corpus of up to 500 long-horizon conversation sessions. The benchmark distinguishes single-session retrieval (facts, preferences, assistant claims) from multi-session and temporally-ordered evidence.

MemoryKG [Suchanek, 2026a] builds a lightweight knowledge graph over verbatim conversation text, where semantic vector search seeds initial candidates and structured graph traversal expands the result set. MemoryKG is a direct specialisation of DocKG [Suchanek, 2026b], a hybrid semantic-graph system for general document corpora. In this paper we report MemoryKG’s performance on LongMemEval and analyse the retrieval characteristics of each question type.

2 The LongMemEval Benchmark

LongMemEval [Wang et al., 2024] is a benchmark for evaluating chat assistants on long-term interactive memory. It contains 500 questions drawn from realistic multi-session conversation histories. Each question is paired with one or more *evidence sessions* — the specific conversation turns or sessions whose content is necessary to answer the question correctly.

The six question types are:

Single-Session User ($n=70$) Questions about facts the user stated in a single session.

Single-Session Assistant ($n=56$) Questions about claims or information the assistant provided in a single session.

Single-Session Preference ($n=30$) Questions about user preferences expressed within a single session.

Knowledge Update ($n=78$) Facts that the user updated or corrected across sessions; the system must retrieve the most recent statement.

Multi-Session ($n=133$) Questions whose answer requires aggregating information from two or more separate sessions.

Temporal Reasoning ($n=133$) Questions that require reasoning about the order or timing of events across sessions.

We use *session-level retrieval recall* as our primary metric: the fraction of evidence sessions whose text is found in the top- k retrieved nodes. We also report NDCG@ k (Normalised Discounted Cumulative Gain), which rewards placing the correct session higher in the ranked list. These are binary retrieval metrics, not end-to-end QA accuracy. The LongMemEval paper reports end-to-end QA accuracy (requiring an LLM to generate an answer); we report whether the evidence session is present in the retrieved context. These are complementary but distinct measures.

3 MemoryKG Architecture

MemoryKG represents a conversation corpus as a four-layer knowledge graph:

1. **Document layer.** Each conversation session is a root document node containing the full session text.
2. **Section layer.** Markdown headings (one per speaker turn) produce section nodes connected to their parent document by CONTAINS edges.
3. **Chunk layer.** Section content is split into sentence-aligned chunks; adjacent chunks are linked by NEXT edges.
4. **Semantic index.** All nodes are embedded with a sentence transformer and stored in a `sqlite-vec` vector store. Retrieval is an exact flat cosine scan rather than an approximate index, so recall at the retrieval stage is exact by construction.

3.1 Query pipeline

A query q is processed in two stages:

Stage 1 — Semantic seeding. The top- k nodes by cosine similarity to the query embedding are retrieved from the vector store, restricted by a metadata prefilter to the sessions in that question’s haystack. These form the *seed set* S .

Stage 2 — Graph expansion. Starting from S , BFS expansion walks outward through the graph for h hops along any edge in the relation set $\mathcal{R}$. The default relation set is

$\mathcal{R} = \{\text{CONTAINS, NEXT, REFERENCES, SIMILAR_TO, HAS_TOPIC, MENTIONS_ENTITY, HAS_KEYWORD}\}$

Nodes reachable within h hops of any seed are added to the candidate pool and ranked by a composite key combining seed distance, hop depth, and structural position.

For LongMemEval evaluations we use $k=10$ seeds and $h=1$ hop.

3.2 Why graph expansion helps for long-horizon retrieval

Consider a Knowledge Update question: “What is the user’s current job title?” The earliest session might read “I just started as a junior analyst,” while a later session reads “Got promoted to senior analyst last month.” A flat vector search that ranks the earlier session higher will return a stale answer. Graph expansion helps via CONTAINS edges: the parent document node for each session contains the full session text, and temporal ordering is encoded in NEXT edges between sessions. When the correct (most recent) session is seeded by any related chunk, expansion propagates recall upward to the session level.

For temporal reasoning questions, the same mechanism resolves ordering ambiguity: if any chunk from the target session appears in the seed set, traversal lifts the entire session into the candidate pool regardless of whether the question vocabulary matches the session’s surface text.

4 Experimental Setup

4.1 Corpus format

For each question, all sessions in the user’s history are written to a temporary directory as Markdown files (one file per session). Each speaker turn becomes a ##-headed section:

```
# Session 2024-03-15

## User

I just got promoted to senior analyst last month.

## Assistant

Congratulations! That’s a big step --- what changed in your role?

## User

More client-facing work, mostly. I’m leading the quarterly reviews now.
```

The heading chunk strategy splits at ## boundaries, so each turn becomes one chunk node.

4.2 Build configuration

- Chunk strategy: **heading** (no per-file embedding at parse time)
- Topic / entity / keyword enrichment: disabled (speed)

- SIMILAR_TO edge discovery: disabled (speed)
- Corpus: 19,195 unique sessions $\rightarrow$ 528,083 nodes, 2,284,295 edges
- Retrieval: $k=50$ seeds, $h=1$, haystack prefilter on, **max_nodes=1000**

The corpus is built once and reused across all 500 questions. Embedding is written to a JSONL cache and the vector index is populated from that cache, so a rebuild of the index costs no model inference.

4.3 Shared embedder

A single `SentenceTransformerEmbedder` (BAAI/bge-small-en-v1.5, 384-dimensional) is initialised once and reused across all questions. This avoids the per-question model reload overhead and is the primary contributor to the 0.68 s/question throughput. The full 500-question evaluation completes in 339.7 seconds on an Apple M5 Max with 64 GB of unified memory.

4.4 Recall and NDCG computation

For each question, let $E = \{e_1, \dots, e_n\}$ be the set of evidence session texts and $N = \{n_1, \dots, n_m\}$ be the ranked list of retrieved node texts. Session-level recall at k is:

$$\text{Recall@}k = \frac{1}{|E|} \sum_{e \in E} \mathbf{1}[\exists n \in N_k : e \subset n \vee n \subset e]$$

where $\subset$ denotes case-insensitive substring containment and N_k is the top- k subset of N . $\text{NDCG@}k$ is computed over binary relevance labels using the standard formulation:

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \quad \text{DCG@}k = \sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)}$$

where $\text{rel}_i \in \{0, 1\}$ indicates whether position i in the ranking contains an evidence session.

5 Results

5.1 Session-level retrieval across k

Table 1 shows MemoryKG’s session-level $\text{Recall@}k$ and $\text{NDCG@}k$ across increasing retrieval depths.

Table 1: MemoryKG session-level retrieval metrics on LongMemEval (500 questions). $\text{Recall}_{\text{any}}$ counts a question as retrieved when at least one gold session appears in the top- k ; $\text{Recall}_{\text{all}}$ requires every gold session to appear. The two differ only for multi-evidence questions.

k	$\text{Recall}_{\text{any}}@k$	$\text{Recall}_{\text{all}}@k$	$\text{NDCG@}k$
1	0.904	0.306	0.904
3	0.972	0.898	0.945
5	0.982	0.968	0.950
10	0.992	0.988	0.954
30	1.000	0.994	0.955
50	1.000	1.000	0.956

Perfect $\text{Recall}_{\text{any}}$ is reached at $k=30$: for all 500 questions, at least one gold session appears in the top-30. The stricter $\text{Recall}_{\text{all}}$ needs $k=50$ to reach 1.000, and the gap between the two

columns is the honest measure of multi-evidence difficulty — at $k=1$ it is the difference between 90.4% and 30.6%, because a single retrieved session cannot satisfy a question whose evidence spans several. By $k=10$ the gap has closed to 0.4 points. NDCG@ k rises from 0.904 at $k=1$ to 0.956 at $k=50$, with the steepest gain between $k=1$ and $k=3$.

5.2 Per-type breakdown

Table 2 shows Recall@5, Recall@10, and NDCG@10 for each LongMemEval question type.

Table 2: MemoryKG retrieval metrics on LongMemEval by question type (top-10, hop=1), recomputed from the per-question results file. Per-type NDCG is omitted: the harness does not emit it, and recomputing it here would apply a different tie-breaking convention from the overall figures in Table 1.

Question Type	n	$R_{\text{any}}@5$	$R_{\text{any}}@10$	$R_{\text{all}}@10$
knowledge-update	78	1.000	1.000	1.000
multi-session	133	0.992	0.992	0.985
single-session-assistant	56	1.000	1.000	1.000
single-session-preference	30	1.000	1.000	1.000
single-session-user	70	0.986	1.000	1.000
temporal-reasoning	133	0.947	0.977	0.970
Overall	500	0.982	0.992	0.988

Four of the six question types achieve 100% Recall@10, and all four reach it on the strict Recall_{all} measure as well. The two types that do not — *temporal-reasoning* (97.7%) and *multi-session* (99.2%) — are those requiring evidence from multiple or temporally ordered sessions. *temporal-reasoning* is also the only type whose Recall@5 falls below 0.98, and it accounts for three of the four misses at $k=10$.

5.3 Comparison with literature baselines

Table 3 places MemoryKG in the context of systems evaluated in the LongMemEval literature [Wang et al., 2024]. Those results are reported as end-to-end QA accuracy, which requires an LLM to generate a free-form answer given retrieved context. MemoryKG’s retrieval recall is a strictly easier task: it measures whether the evidence is *present* in the retrieved context, not whether a reader can extract the correct answer from it. The two metrics are complementary but not directly comparable; the table is included for broader orientation.

Table 3: Context comparison. † QA accuracy from Wang et al. [2024] (requires LLM answer generation); ours is retrieval recall (no LLM). Different metrics; not directly comparable.

System	Score	Metric	LLM required
MemoryKG (ours, R@10)	99.2%	Retrieval recall	None
MemoryKG (ours, R@30)	100%	Retrieval recall	None
GPT-4o full-context†	~66%	QA accuracy	Yes
GPT-4o + RAG†	~50%	QA accuracy	Yes
GPT-4-Turbo full-context†	~56%	QA accuracy	Yes
Claude-3.5-Sonnet†	~52%	QA accuracy	Yes
ReadAgent†	~42%	QA accuracy	Yes
Zero-shot (no memory)†	~25%	QA accuracy	Yes

5.4 Miss analysis

At $k=10$, exactly 4 out of 500 questions returned no evidence session in the top-10 retrieved nodes. These four misses correspond to question IDs 10d9b85a, gpt4_468eb064, eac54add, and gpt4_68e94288. All four were recovered by $k=30$, confirming that the evidence exists in the index but ranks below position 10 in these cases. The miss rate at $k=10$ is 0.8% (4/500). Three of the four are *temporal-reasoning* questions and one is *multi-session*; no other question type misses at all.

6 Analysis

6.1 Why temporal-reasoning is the hardest type

Temporal-reasoning questions ask about the ordering or recency of events across sessions (e.g., “When did the user first mention they were learning Spanish?”). The evidence session may contain only a passing reference to the target event, while semantically closer sessions discuss unrelated topics. Vocabulary mismatch between the question and the target session is highest in this category: a question about “first mention” may correspond to a session that simply reads “I started a Spanish class” without temporal markers. Graph expansion partially compensates — if any chunk from the same session appears in the seed set, NEXT and CONTAINS edges recover the full session — but 3 out of 133 temporal-reasoning questions (2.3%) still miss at $k=10$.

6.2 Why multi-session questions remain hard

Multi-session questions require evidence from two or more sessions simultaneously. At $k=10$, the system must place all required sessions within the top-10 result, which becomes more constrained as the number of required sessions grows. Unlike the ConvoMem tier model [Pakhomov et al., 2025] (which fixes the number of evidence messages), LongMemEval multi-session questions vary in evidence count. The 1 miss at $k=10$ for this category represents a case where one required session ranked below position 10 while the other(s) ranked highly.

6.3 The NDCG plateau and its implications

$\text{NDCG}@k$ rises sharply from 0.904 at $k=1$ to 0.945 at $k=3$, then climbs only slowly — 0.950, 0.954, 0.955, 0.956 at $k=5, 10, 30, 50$. The near-flatness beyond $k=3$ indicates that additional retrieved nodes do not substantially improve the rank of correct sessions: they are already near the top of the list when present. The residual gap to 1.0 reflects the four hard misses that surface only beyond position 10, plus the multi-evidence questions whose second and third gold sessions rank lower than their first. For a downstream reader consuming the top- k retrieved context, $k=10$ is effectively optimal: it captures 99.2% of questions and 98.8% on the strict all-evidence measure, and further depth buys 0.002 of NDCG.

6.4 Throughput and scaling

At 0.68 s/question on Apple Silicon (M5 Max, 64 GB), MemoryKG answers 500 questions in 339.7 seconds. Indexing is not part of that figure. The corpus is compiled once — 19,195 sessions into 528,083 nodes — and every question is then served from the same prebuilt store, so per-question cost is a vector scan plus one hop of graph expansion, not an ingestion. The higher per-question time than ConvoMem (0.68 s vs. 0.07 s) follows from corpus size: the exact cosine scan is linear in the number of indexed vectors, and LongMemEval’s index is roughly two orders of magnitude larger. This is the deliberate trade the architecture makes — an exact scan costs more per query than an approximate index but removes recall loss at the retrieval stage, which is what makes the numbers above reproducible.

7 Conclusion

MemoryKG achieves 99.2% Recall@10 and 100% Recall@30 on the 500-question LongMemEval benchmark, with an NDCG@10 of 0.954 and no LLM required at any stage. Four of six question types reach perfect Recall@10, on the strict all-evidence measure as well as the any-evidence one. The two remaining types — temporal-reasoning and multi-session — reflect the fundamental difficulty of retrieving evidence from multi-session contexts with temporal ordering requirements; both approach but do not reach 100% at $k=10$, and both are fully resolved by $k=30$.

The hybrid architecture — semantic seeding combined with structured graph expansion — enables high-precision retrieval without the latency or cost of LLM inference. The near-flat NDCG from $k=3$ onward (0.945 to 0.956 across a sixteen-fold increase in k) confirms that when evidence sessions are retrieved, they rank near the top of the result list, making the system suitable for downstream readers with tight context budgets.

Limitations. Recall measures whether the evidence session is *present* in the retrieved context; it does not measure whether a downstream reader can correctly extract the answer. End-to-end QA accuracy on LongMemEval with a MemoryKG retrieval front-end is left for future work. The 4 misses at $k=10$ are all resolved by $k=30$ and are concentrated in the temporal-reasoning category (3 of 4), suggesting that temporal ordering cues are the primary remaining failure mode.

A further limitation is comparative rather than technical. Against MemoryPalace hybrid v2 — the strongest LLM-free configuration published for this benchmark — MemoryKG leads on Recall@10 (99.2% vs. 99.0%) and NDCG@10 (0.954 vs. 0.934) but trails at Recall@5 (98.2% vs. 98.4%), and MemoryPalace’s best overall result still requires a Haiku reranking pass. Recall@1, at 90.4%, is this system’s weakest metric: the graph places the right session in the top ten almost always, and first less reliably than a tuned dense retriever.

Reproducibility. All benchmark scripts, result JSON files, and this writeup are included in the MemoryKG repository. The full evaluation can be reproduced with:

```
# Build once: parse -> SQLite, embed -> JSONL cache, index from cache
python benchmarks/longmemeval/longmemeval_memkg.py prepare \
    benchmarks/longmemeval/data/longmemeval_s_cleaned.json \
    --wipe --chunk-strategy heading

# Evaluate (haystack filter, k=50, hop=1 are defaults)
python benchmarks/longmemeval/longmemeval_memkg.py run \
    benchmarks/longmemeval/data/longmemeval_s_cleaned.json \
    --out benchmarks/longmemeval/results_20260826_bge_haystack.jsonl
```

The per-question results backing every table above are checked in at `benchmarks/longmemeval/results_20260826`

References

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